

Ez-Router Pendant



Joystick

Axis Motion – Jogging

The joystick is a spring centered mini-joystick that moves multiple axis. (the rotary switch must be in either the X/Y or Z/A position)

Joystick Pushbutton – Enable/Disable

- When the red LED in “Micro” is slowly flashing, the Pendant is disabled. Pushbuttons and joystick motion will not function, except for the **Joystick Pushbutton**.
- Pressing the **Joystick Pushbutton** and holding it down for 1 second will enable all of the pushbuttons and joystick motion and turn the Enable/Disable LED a solid green.
- When there has been 45 seconds of inactivity using the Pendant, a quick flashing red Enable/Disable LED will warn that there is 15 seconds left until the Pendant is disabled.
- Any activity (joystick or pushbutton) will reset the timer to 1 minute and return the Enable/Disable LED to a solid green.
- At the end of the “15 second warning”, the Pendant will be disabled and the Enable/Disable LED will flash red
- The stop button is always active even when the Pendant Enable/Disable led is flashing red.

Joystick Pushbutton - Mode Selection

- There are 2 green “mode” LEDs above the Joystick. Pressing the **Joystick Pushbutton** changes the mode.
- When both LEDs are on, the joystick is in **Normal (Norm)** speed mode.
- When only the left green LED is on, the joystick is in **Incremental (Incr.)** speed mode. When operating the X/Y/Z/A axes, this is the same as holding down the “Ctrl” key on the keyboard before operating the arrow keys. Joystick must return to center for each incremental axis movement in **Incremental** speed mode.
- When only the right green LED is on, the joystick is in **Fast** speed mode. When operating the X/Y/Z/A axes, this is the same as holding down the “Shift” key on the keyboard before operating the arrow keys.

Pushbuttons

Start

- Same as “Cycle Start” or <Alt-R> from keyboard.

Pause

- Same as “Feed Hold” or <Space> from keyboard.

Stop

- Same as <Alt-S> from keyboard. **Stop is ALWAYS enabled. (NOT an ESTOP).**

Goto

- Initiates action when the **Selector Switch** is set to “Zero”, **SafeZ**, **ToolC**, **RefAll**, or **Rewind**.

Mac1

- Initiates Auto Z setter function when the **Selector Switch** is set to **Macro**.

Mac2

- Initiates Regenerate Tool Path function when the **Selector Switch** is set to **Macro**.

Mac3

- Initiates Vacuum Bed On/Off function when the **Selector Switch** is set to **Macro**.

Z1/Tog

- Zeros X DRO when the **Selector Switch** is set to **X/Y**.
- Zeros Z DRO when the **Selector Switch** is set to **Z/A**.
- Toggles Spindle (CW) On/Off when the **Selector Switch** is set to **SOvr**.

Z2/Res

- Zeros Y DRO when the **Selector Switch** is set to **X/Y**.
- Zeros Aux DRO when the **Selector Switch** is set to **Z/A**.
- Resets Feed Override when the **Selector Switch** is set to **FOvr**.
- Resets Spindle Override when the **Selector Switch** is set to **SOvr**.

Selector Switch

- **Joystick=X/Y**
- Joystick is controlling the X/Y axis.
- **Joystick=Z/A**
- Joystick is controlling the Z/A axes.

“Zero”

- Pressing **“Goto”** moves **X, Y, Z,** and **A** axes to their “zero” positions per Mach3 DROs.

SafeZ

- Pressing **“Goto”** moves Z axis to the **“Safe Z”** position configured in Mach3.

ToolC

- Pressing **“Goto”** moves the spindle to the **“Tool Change”** position configured in Mach3.

RefAll

- Pressing **“Goto”** references the machine (same as RefAllHome)

Rewind

- Pressing **“Goto”** “rewinds” the G-code file loaded in Mach to the beginning.

Macro

- Pressing **“Mac1”** initiates the Auto Z-Setter Function.
- Pressing **“Mac2”** initiates the Regenerate Tool Path function.
- Pressing **“Mac3”** initiates the Vacuum Bed On/Off function.

X/Y

- Moving the joystick jogs the X and or the Y axis.
- Pressing **“Z1”** zeros X DRO.
- Pressing **“Z2”** zeros Y DRO.

Z/A

- Moving the joystick jogs the Z and or the A axis.
- Pressing **“Z1”** zeros Z DRO.
- Pressing **“Z2”** zeros A DRO.

SOvr (Spindle Speed Override)

- Joystick Forward/Backwards movement increases/decreases spindle speed
- Pressing **“Z1/Tog”** toggles spindle On/Off.
- Pressing **“Z2/Res”** resets the spindle override.

FOvr (Feed Rate Override)

- Joystick Forward/Backwards movement increases/decreases feed rate.
- Pressing “**Z2/Res**” resets the feedrate override.

Notes:

- Always use the PS2 keyboard to USB adapter **supplied** with the Pendant for proper operation.
- Although a keyboard cable as long as 250 feet has been used successfully with the pendant, a 25 foot cable is all that is supported.
- Unless the 4th axis (“Aux” or “A”) is enabled in Mach, no action will take place when trying to operate the “A” axis with the joystick or buttons using the Pendant.